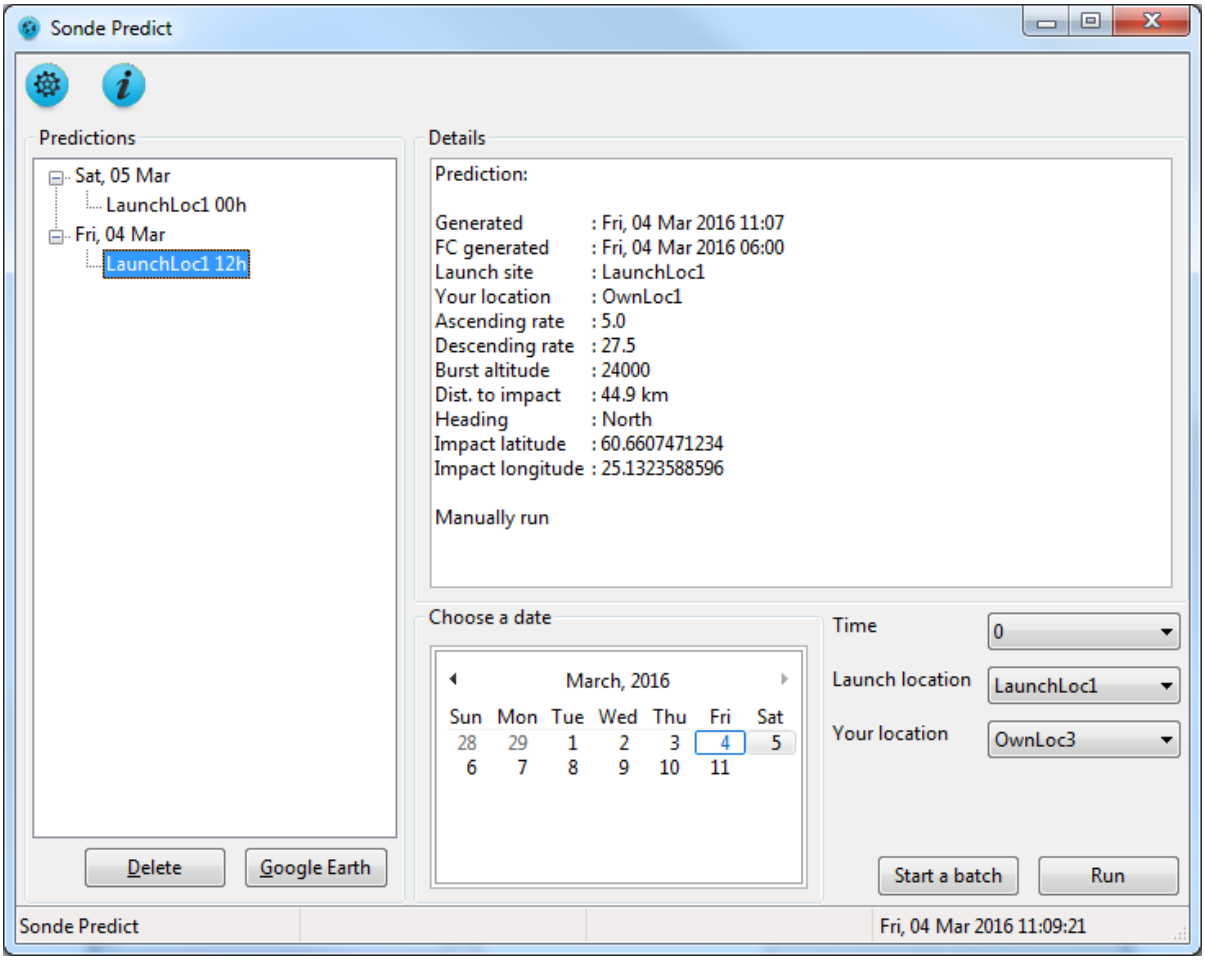
# Sonde Predict user manual

## 1. Description

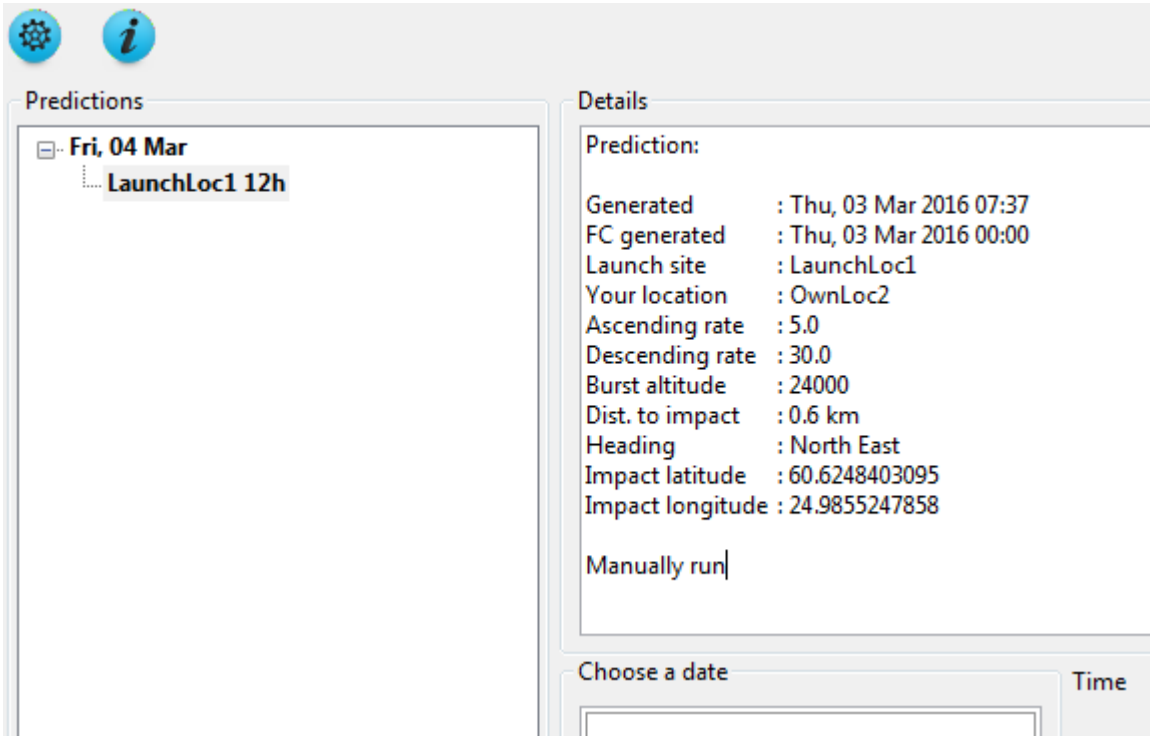
Sonde Predict is software for calculating radiosonde flight paths using upper wind data gathered from Global Forecast System (GFS). It automatically downloads all needed parameters from GRIB-files and then uses, a highly sophisticated, algorithm to estimate the flight path and impact position. The user interface is designed to be simple and user friendly. Users can define multiple launch and receiving sites and run multiple calculations also in batched mode. Results are displayed numerically or using 3D-track presentation in Google Earth.

## 2. Generic information about software

The user interface has been designed to be very self-controlling and simple to use. All calculated prediction dates are displayed on the left. Prediction results can be displayed by selecting date and time. The details window display all needed information including estimated impact location coordinates. The coordinates are decimal formatted.



Predictions date(s) will be **bold** when the estimated impact position is under 30 km from configured receiving location (Your location). This will make easier to detect good radiosonde hunting days quickly without analyzing data more precisely.




Prediction date is entered using Schedule-dialog and time from dropdown menu. It's possible to create history calculations also from previous 7-days timeframe.

Prediction is started by clicking Run or Start a batch button. Batch button will do multiple calculations (configured under settings) automatically.

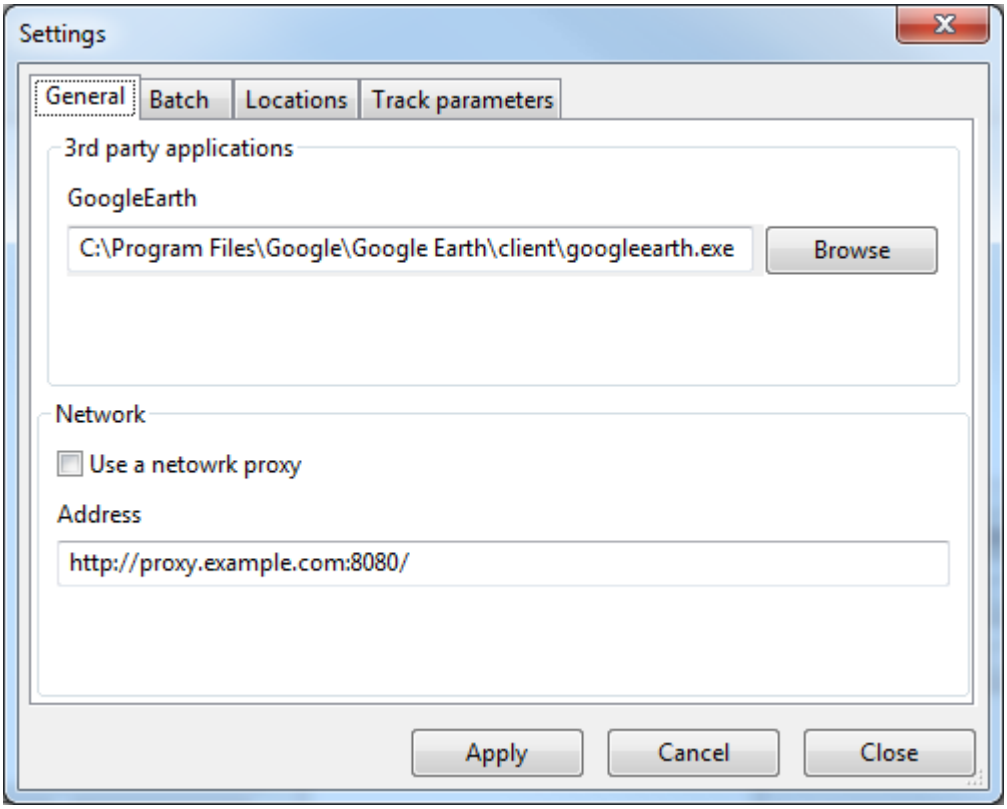
Already calculated predictions can be deleted by selecting date(s) and clicking "Delete" button.

To visualize track using Google-Earth, select one or more dates keeping Ctrl-button pressed and then click Google Earth button. More about Google Earth integration later in this document.

## 3. Settings

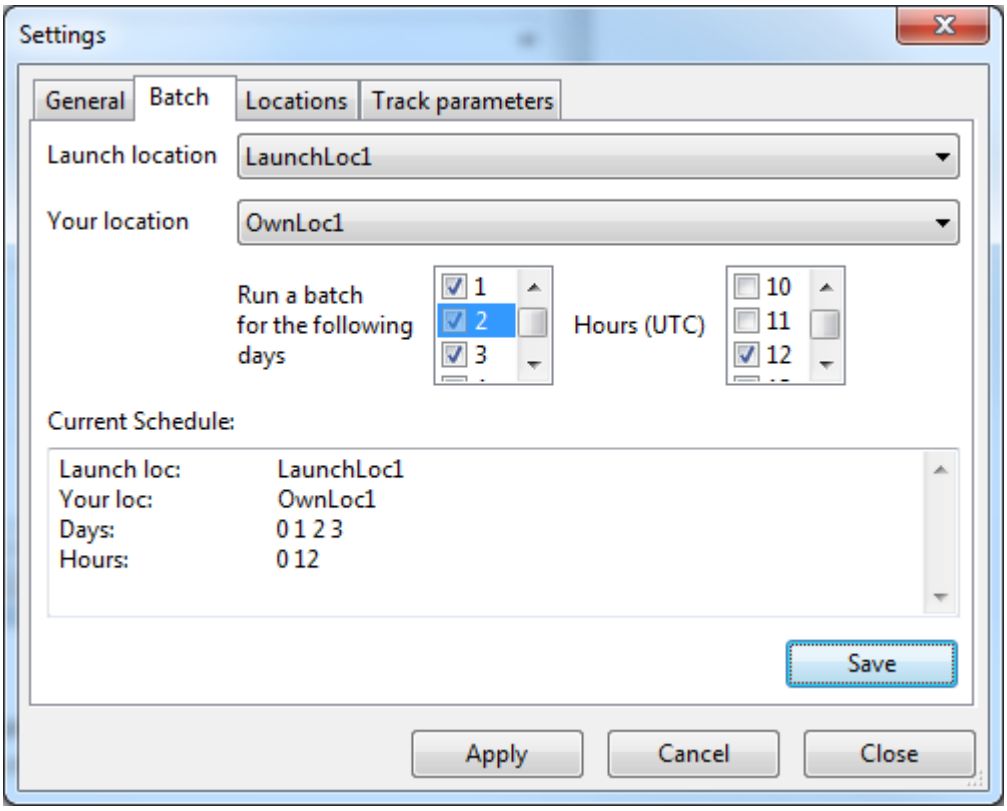
Click  button to open program settings window.

To allow Google-Earth 3D-track visualization, select correct path to googleearth.exe executable.




If specific network proxy is needed it can be set under Network section.

Batch predictions are useful to save manual work if calculations for multiple different times are needed. A typical situation is that user want to calculate 00 UTC & 12 UTC soundings for next few days.

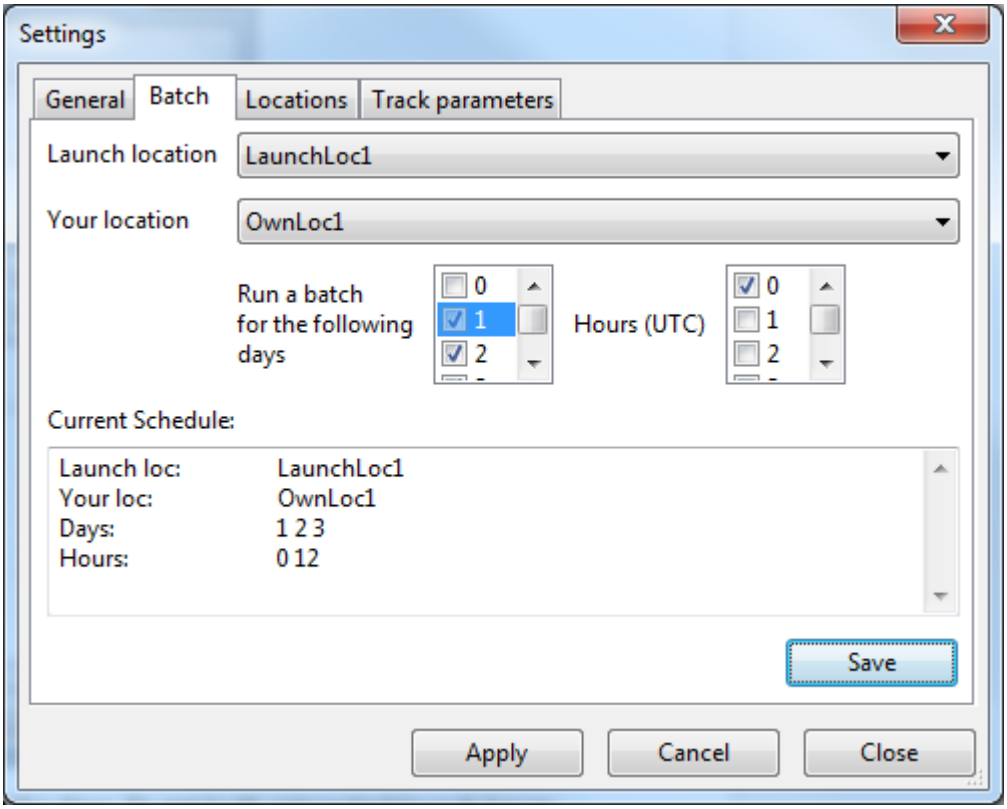



To create new batch rule, select launch site (Launch location) and your own location (Your location) from dropdown menus. Then select how many days are included into calculations. Days numbering: 0 = Current day, 1 = tomorrow . . .

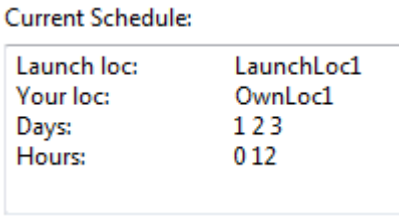
Finally select needed launch time hours. Click Save and recheck created schedule and save changes by clicking Apply.

**Example**

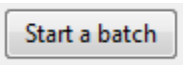
User want to create batch prediction for the next three days 00 and 12 UTC flights excluding the current day.




Select launch location and your location. Then select 1, 2 and 3 for days. To get both interested launch hours to be included into prediction, select 0 and 12. Finally click Save to activate batch schedule.



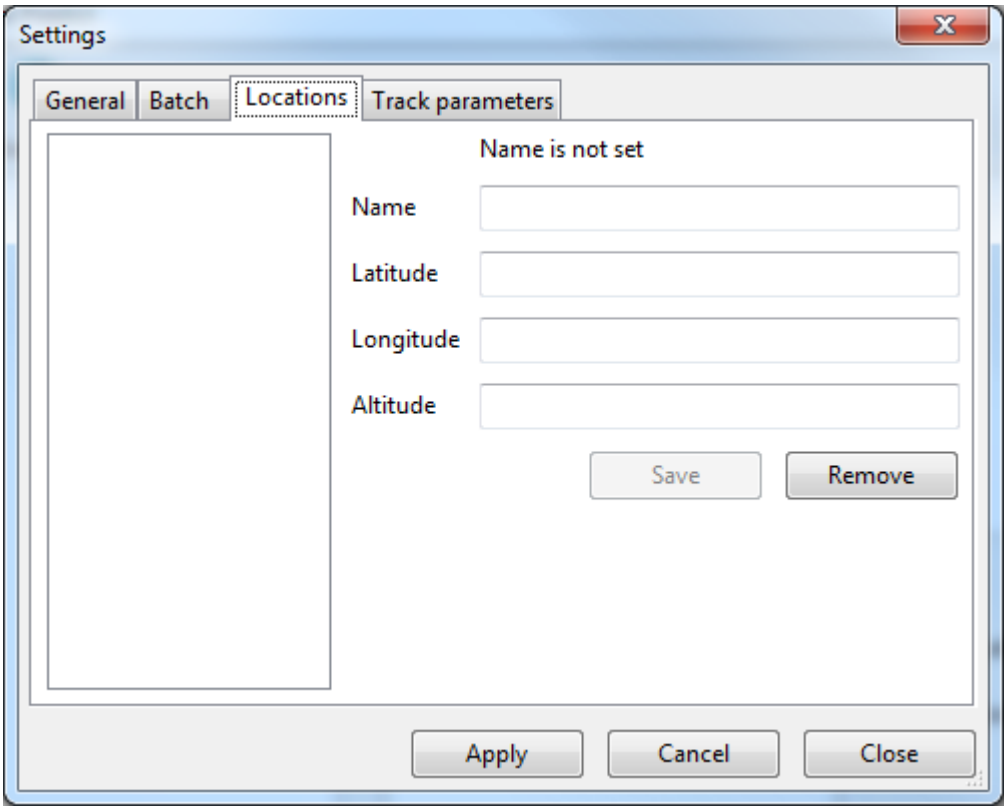

When schedule is activated it can be cross-checked from the Current Schedule dialog. Days 1 2 3 mean that next three days are included into prediction, but calculation for current day (0) is not included. Hours 0 12 indicate that 00 UTC and 12 UTC hours are calculated for each selected days.

When batch schedule is created it can be anytime executed by pressing 

button.

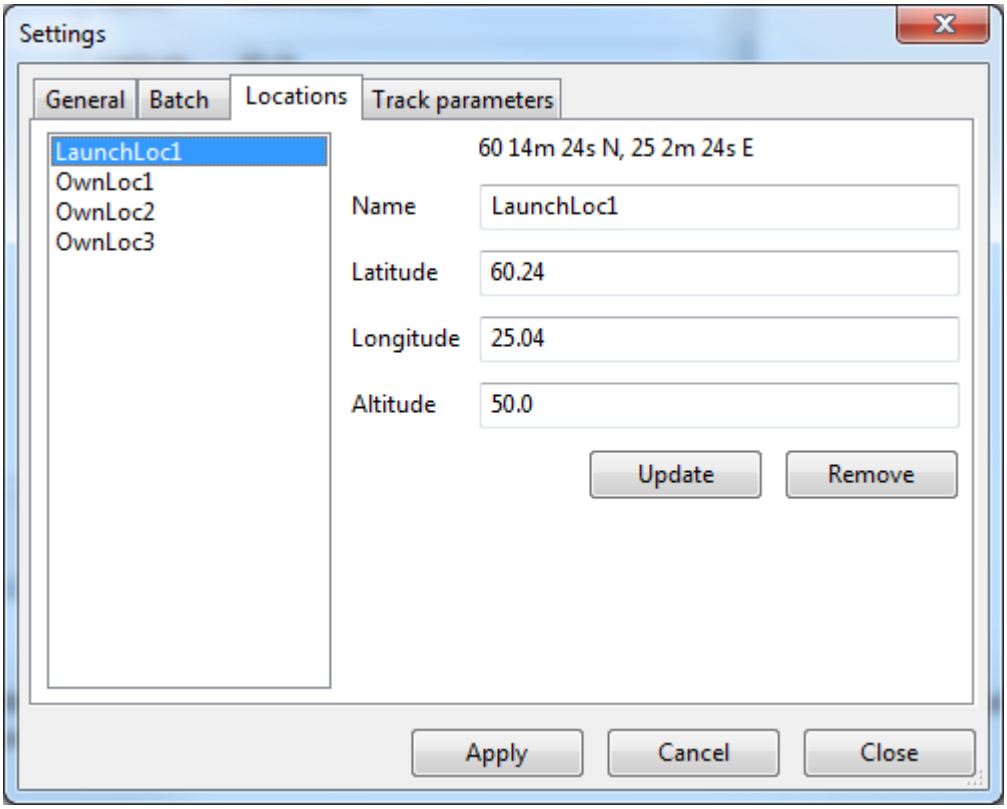
Note: Locations are set under Locations-tab. Multiple different locations can be set and they are then easily available via dropdown menus.

When program is started first time, there is no any locations available.




To add one, specify name, latitude, longitude (negative value for western locations) and station altitude in meters. Finally click Save to add new location. Note: Coordinates are entered as decimals.

Locations dialog seen after few launch and own locations added.

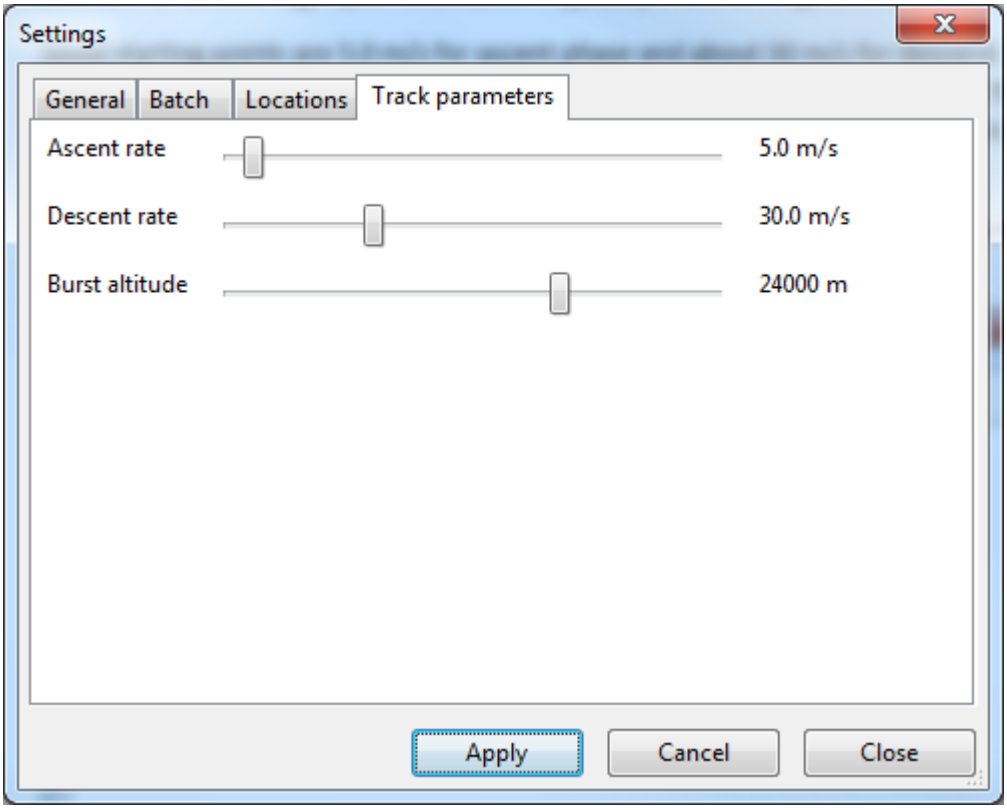



To update already added location select interested one from the list, change needed values and click Update. Already added location can be deleted by selecting location name and then click Remove.

Balloon flight related parameters are configured via Track parameters tab. Proper selected settings is the key element of reliable flight path estimation. Typically these settings are based of some kind of experimentation.

Good starting point is 5.0 m/s for ascent phase and about 30 m/s for descend. Descend rate is averaged value of overall fall phase of flight. On upper atmosphere levels, it is higher and then when density of air increases at lower altitudes during descend it begins to decrease. Using the data gathered from actual soundings would help a lot finding the best values.

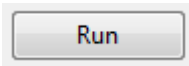
Burst altitude is estimated balloon burst level in meters. This setting will significantly affect estimated impact point.



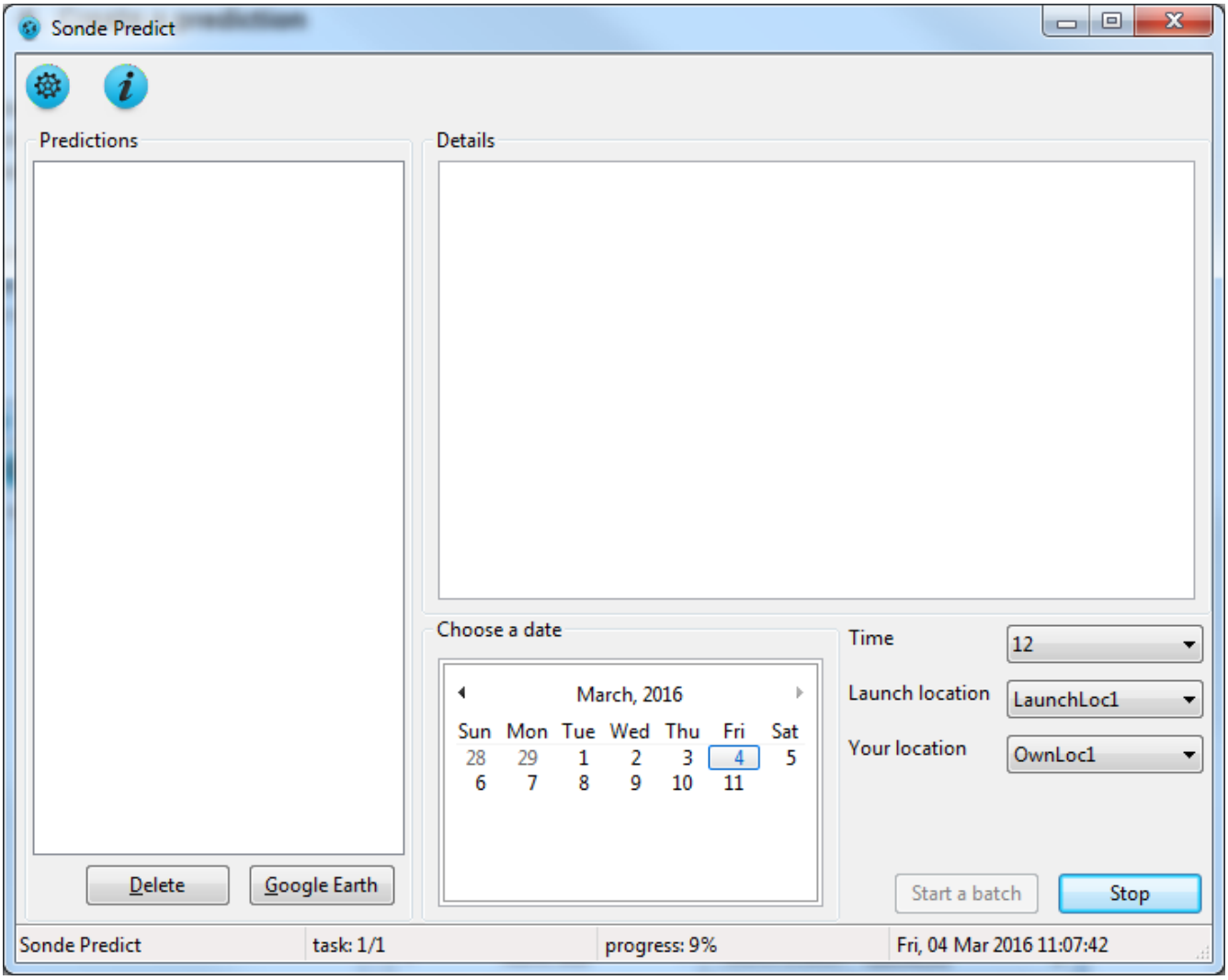

Settings can be set by sliders using mouse and fine tune is done by using keyboard left/right arrow buttons. When all needed settings are done click Apply and Close.

## 4. Create a prediction

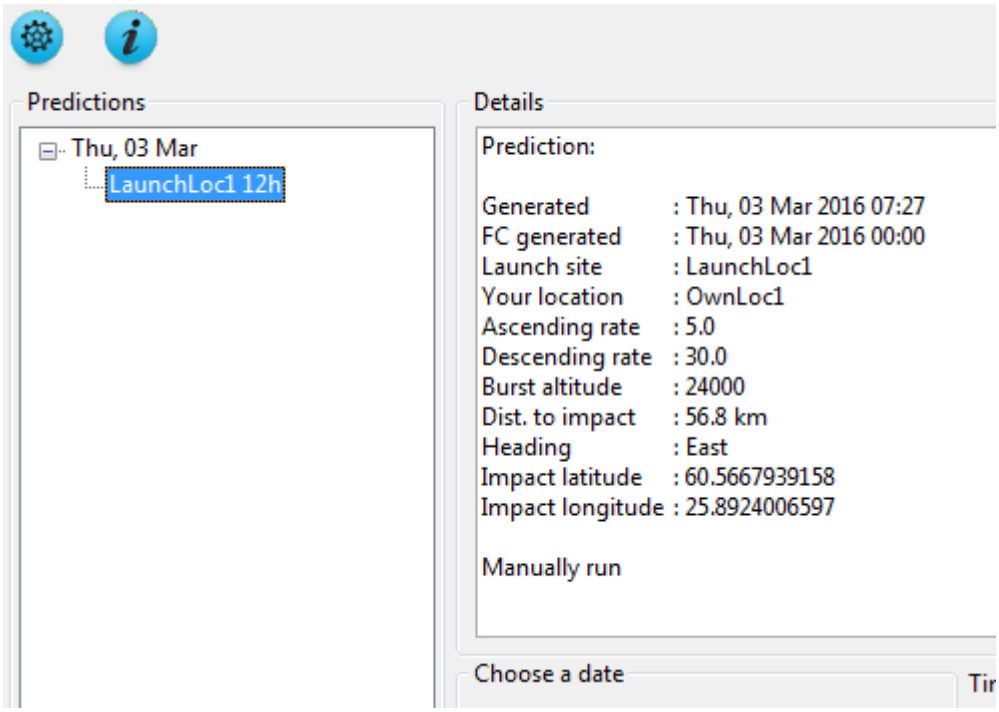
Select balloon launch location and your own location. Then select launch date from the calendar and UTC-time from the dropdown menu. Remember to add all needed locations via Settings → Locations, before running predictions.

Start prediction by pressing 

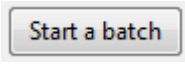
button. Progress percentage will inform the phase of calculation. If there is any errors during processing they are displayed via separate error dialogs. See troubleshooting section for more info. Calculation can be cancelled anytime by pressing Stop button.



When calculation is ready, needed data is available by clicking date and plus-sign.




TIP: It's possible to select needed data fields by mouse and then just copy them to clipboard using Ctrl+c keyboard command.

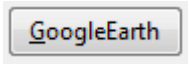
Batch run is done similar way by pressing 

button. Calculation time will vary depending how many days and hours are selected. To monitor phase of the batch calculation see task and progress information fields on the bottom of the main Sonde Predict window. Progress will always indicate single task phase not the state of the whole batch prediction.

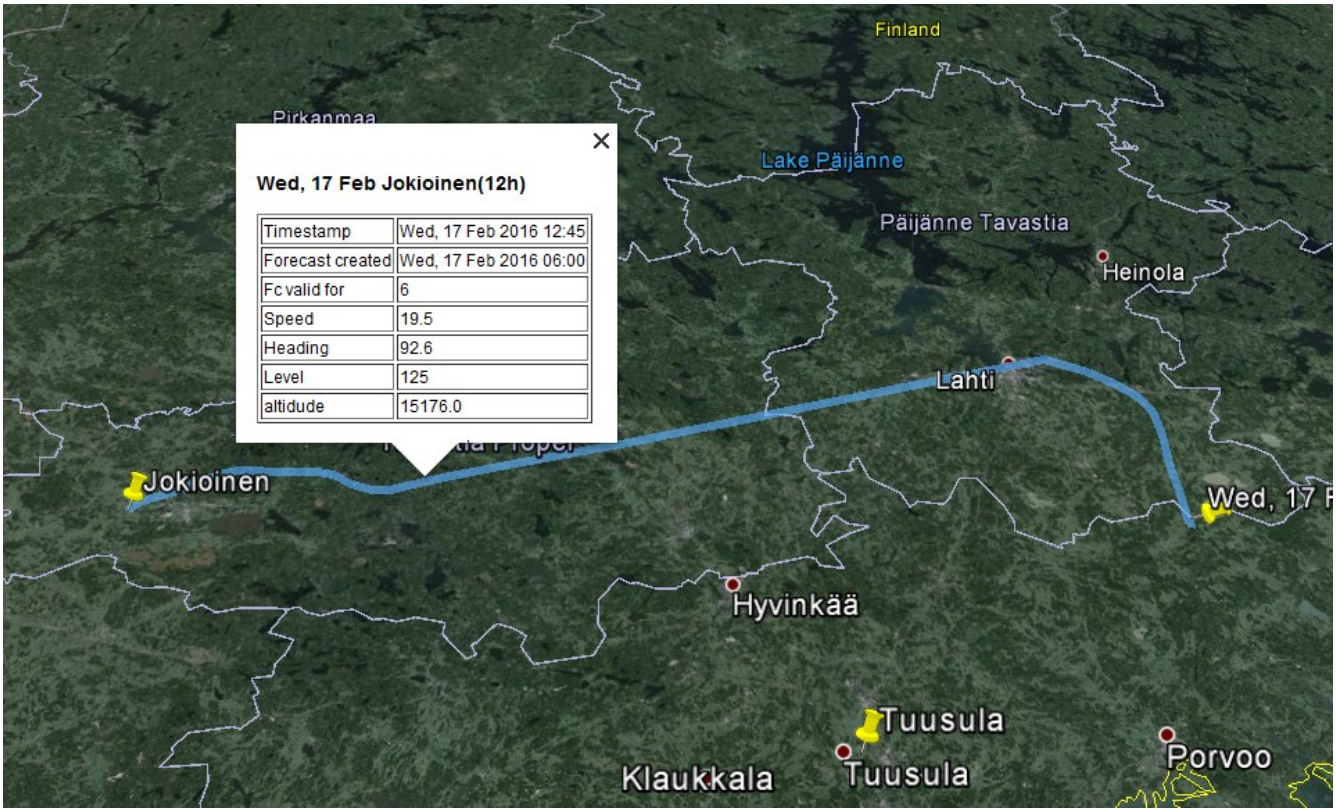



If prediction is made again for the same locations using same date and time, current prediction results will be overwritten automatically. Also if new calculation is done inside small timeframe, data is taken from the program cache to make operations faster.

## 5. Google-Earth visualization

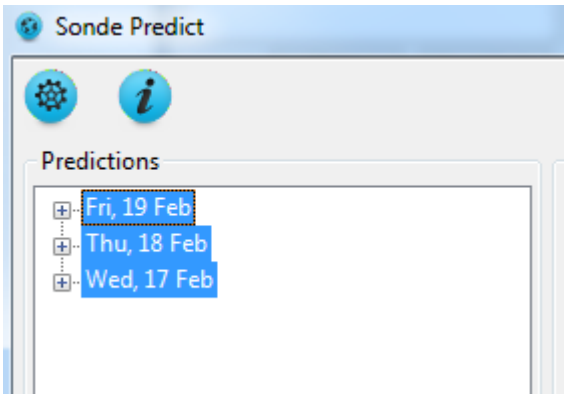
Sonde Predict has the capability to visualize calculated paths using GE. To create simple visualization from one flight select date and then press 

button to export .kml file to GE. Google-Earth should automatically start and display track after a short amount of time.

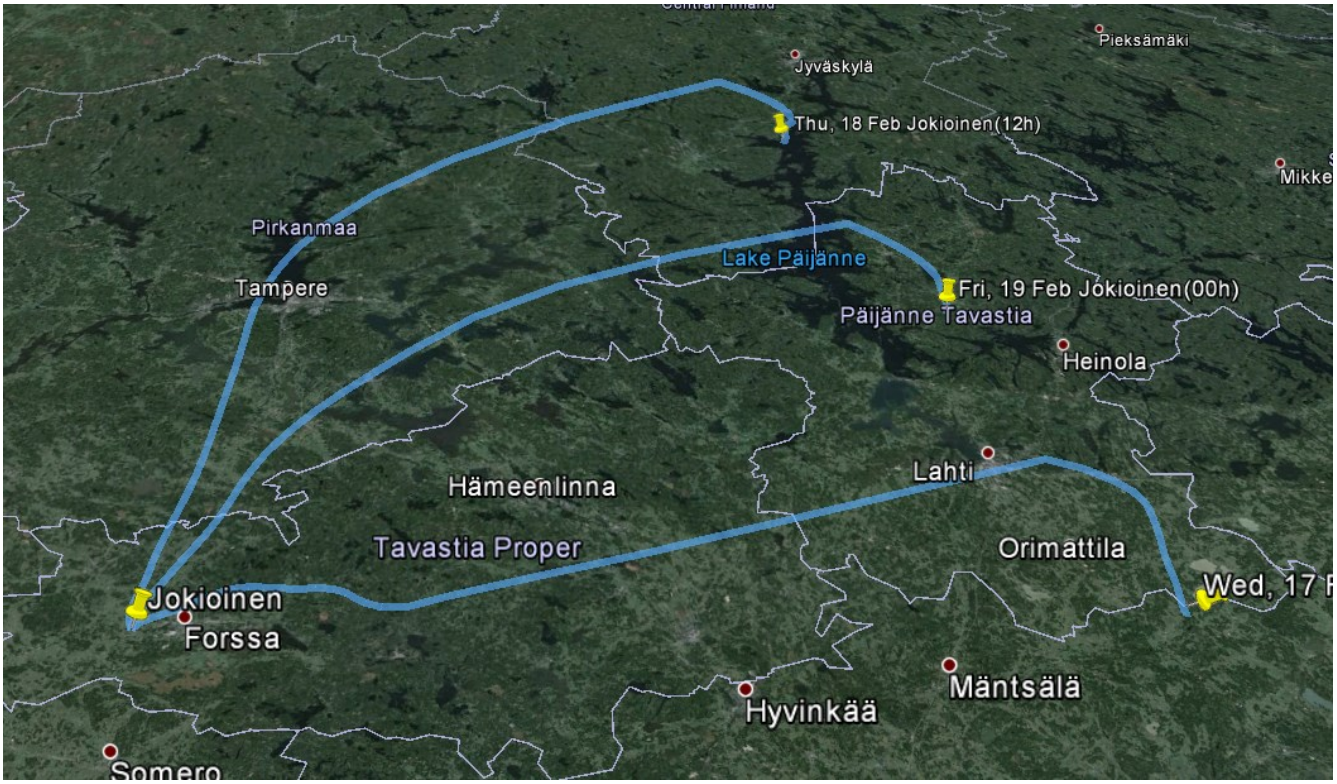



Additional information regarding different phases of estimated flight can be gathered by clicking the track line by mouse cursor. Forecasted wind speed and heading at selected altitude are displayed in separate data-dialog. This feature can be also to be used for crosschecking valid forecast time used for calculation.

Multiple tracks can be also displayed at the same time. This is done by selecting multiple dates keeping Ctrl-button pressed and then clicking GoogleEarth button.




Tracks are then displayed at the same time on GE to give user better knowledge about changed upper atmosphere wind speeds and directions. Note: When selecting multiple days, export time to Google Earth would be little bit longer comparing the single one.

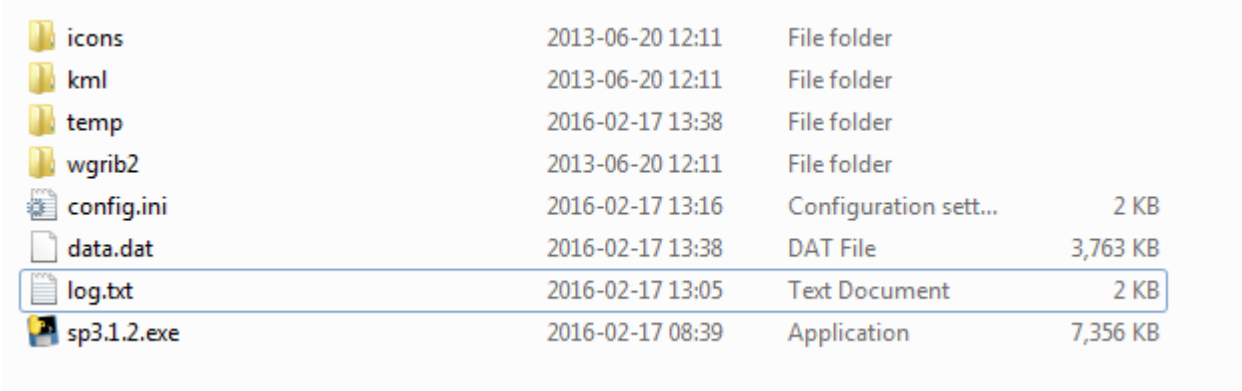



Additional info-dialog is available normally also when multiple tracks are displayed.

## 6. Troubleshooting

Typical error situations are related to networking. If valid internet connection is not available, downloading needed GFS-data will fail and produce errors. In these cases please recheck your network settings and connectivity.

Separate logfile log.txt is automatically created into Sonde Predicts folder. Please refer this file in error situations.



| | | | |
|---|---|---|---|
| icons | 2013-06-20 12:11 | File folder | |
| kml | 2013-06-20 12:11 | File folder | |
| temp | 2016-02-17 13:38 | File folder | |
| wgrib2 | 2013-06-20 12:11 | File folder | |
| config.ini | 2016-02-17 13:16 | Configuration sett... | 2 KB |
| data.dat | 2016-02-17 13:38 | DAT File | 3,763 KB |
| log.txt | 2016-02-17 13:05 | Text Document | 2 KB |
| sp3.1.2.exe | 2016-02-17 08:39 | Application | 7,356 KB |

## 7. Installation

There is no need to run any installer. Just create new folder ie.  C:\SondePredict and unzip needed files into folder created.

Sonde predict is started by double clicking **sp.exe** file. User can create shortcut to Windows desktop if necessary.

Supported Operating Systems:

- Windows XP SP3
- Windows 7 32-bit & 64-b
- Windows 10

To uninstall Sonde Predict, just remove folder and files. Sonde Predict does not write any data to the windows registers.